SSEE and the Hubble Tension: An Algebraic Local Screening Fraction

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Abstract

The Hubble tension between the CMB-inferred value $H_0^{\rm CMB} = 67.4 \pm 0.5$ km/s/Mpc and the local distance-ladder measurement $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc remains one of the sharpest anomalies in modern cosmology. Within the Structural Self-Energy Expansion (SSEE) framework, the algebraic constants φ (golden ratio) and π determine all parameters with zero free parameters. We derive algebraically that the dark-energy EFT of Paper 7 and the disformal geodesic of Paper 8 together imply a *local screening fraction*

$$f_{\rm screen} \;=\; \frac{\alpha_K}{3\,\mathcal{M}} \;=\; \frac{\pi-\varphi}{(\varphi+\pi)^2} \;\approx\; 0.06725\,,$$

where $\alpha_K = 3\,\Omega_{\rm DE}(1+w_0)$ is the EFT kineticity derived from the background equation of state, and $\mathcal{M} = \beta_c/2 = (3\varphi+\pi)/4 \approx 1.9989$ is the effective optical coupling derived from the disformal null geodesic. The two structural constants $\mathcal{A} = 2\mathcal{M}$ cancel exactly in the ratio, leaving a pure function of φ and π. Applied as a local correction to the global background anchor $H_0 = 3(\varphi+\pi)^2 = 67.962\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (Postulate D, Paper 1),

$$H_0^{\rm local} \;=\; \frac{H_0}{1-f_{\rm screen}} \;=\; \frac{67.962}{1-0.06725} \;\approx\; 72.86 \text{ km/s/Mpc}\,,$$

which lies 0.17σ from SH0ES (Riess et al. 2022) with *zero free parameters*. This adopts the single global anchor $H_0 = 3(\varphi+\pi)^2$ shared by every sector of the suite (2026 reframe). The earlier register used the CMB-mapped value $H_0^{\rm MIRA} = 67.037$, giving $H_0^{\rm local} = 71.87$ (1.12σ); this is now superseded. The 2026 ω_m-direct reframe adopts the single algebraic anchor $H_0^{\rm alg} = 3(\varphi+\pi)^2 = 67.962$ as the global background shared by all sectors — with *no* matter factor, the CMB density following as $\Omega_{m,\rm CMB} = \omega_m/h^2$ (Paper 1). The correction operates at late times ($z < 0.15$) through the scalar kinetic energy contribution to the local expansion rate, and does not modify the sound horizon r_d, the CMB peak positions, or the growth factor σ_8 derived in Papers 3 and 5. The UV completion of Paper 10 gives $H_0^{\rm UV} = 73.040\,$km/s/Mpc (0.00σ from SH0ES), the small-scale refinement of the same cascade.

Contents

1 Introduction

The Hubble tension [Verde et al., 2019, Riess et al., 2022, Kamionkowski & Riess, 2023, Di Valentino et al., 2021, Knox & Millea, 2020] is the 4–6σ discrepancy between measurements of the present expansion rate obtained by two independent methods:

- **Early-universe inference**: CMB anisotropies (Planck PR4, Planck Collaboration 2020) combined with acoustic oscillation physics yield $H_0 = 67.4 \pm 0.5$ km/s/Mpc under the assumption of ΛCDM.

- **Local distance-ladder**: Type Ia supernovae [Riess et al., 1998, Perlmutter et al., 1999] calibrated with Cepheids in the SH0ES programme [Riess et al., 2019, 2022] give $H_0 = 73.04 \pm 1.04$ km/s/Mpc, probing predominantly $z < 0.15$.

The SSEE framework [Almeida, Paper1,P,P,P] derives the cosmological expansion rate algebraically:

$$H_0^{\mathrm{alg}} \;=\; 3(\varphi + \pi)^2 \;\approx\; 67.96\ \mathrm{km/s/Mpc}\,, \tag{1}$$

which agrees with Planck at the 1.1σ level [Almeida, Paper2] but does not, at first sight, resolve the local tension.

In this paper we show that the same algebraic structure that produces H_0^{alg} also produces a local kinematic correction $f_{\mathrm{screen}} \approx 0.06725$ from the SSEE algebra. Applied to the global background anchor $H_0 = 3(\varphi + \pi)^2 = 67.962$ km/s/Mpc, this predicts $H_0^{\mathrm{local}} = 72.86$ km/s/Mpc (0.17σ from SH0ES). The derivation combines two results already established in this series:

1. From Paper 7 [Almeida, Paper7]: the EFT kineticity $\alpha_K = 3\,\Omega_{\mathrm{DE}}(1 + w_0)$, which follows from the background Friedmann equations and the k-essence action $K(X) = X/\mathrm{KAL}_0$.

2. From Paper 8 [Almeida, Paper8]: the effective optical coupling $\mathcal{M} = \beta_c/2 = (3\varphi + \pi)/4$, which emerges from the disformal null geodesic of dark matter.

Together these give the exact algebraic identity $f_{\mathrm{screen}} = \alpha_K/(3\mathcal{M})$.

Organisation. Section 2 recalls the key results from Papers 7 and 8. Section 3 derives the algebraic identity in three steps. Section 4 interprets the result physically. Section 5 verifies consistency with the rest of the SSEE series. Section 6 states the pre-committed falsifiable prediction and its current observational status. Section 7 compares the mechanism quantitatively against early dark energy, self-interacting dark radiation, and late dark-energy transitions. Section 9 summarises.

2 Background: EFT Kineticity and Disformal Coupling

2.1 The SSEE action (Paper 7)

The fundamental action of SSEE is a k-essence scalar [Armendáriz-Picón et al., 1999, 2001] minimally coupled to gravity and conformally coupled to dark matter:

$$S = \int d^4x\,\sqrt{-g}\,\left[\frac{M_{\mathrm{Pl}}^2}{2}R + K(X,\phi) + \mathcal{L}_m\right]\,, \tag{2}$$

with kinetic kernel

$$K(X,\phi) = \frac{X}{\mathrm{KAL}_0} + \frac{X^2}{M^4} - V_0\, e^{-\lambda\phi/M_{\rm Pl}}\,, \tag{3}$$

where $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ and

$$\mathrm{KAL}_0 = \frac{\varphi+\pi}{2} + \pi \approx 5.521\,, \tag{4}$$

$$\beta_c \equiv \mathcal{A} = \frac{3\varphi+\pi}{2} \approx 3.998\,. \tag{5}$$

All parameters are algebraically fixed; no fitting is performed.

The dark-energy equation-of-state parameters derived in Paper 1 are

$$w_0 = -\frac{\mathcal{A}}{\varphi+\pi} \approx -0.840\,, \tag{6}$$

$$\Omega_{\rm DE} = -w_0 = \frac{\mathcal{A}}{\varphi+\pi} \approx 0.840\,. \tag{7}$$

The EFT kineticity in the Bellini–Sawicki classification [Bellini & Sawicki, 2015], part of the broader effective field theory of dark energy [Gubitosi et al., 2013, Gleyzes et al., 2013], is defined as

$$\alpha_K \;\equiv\; \frac{2X\,K_X}{M_{\rm Pl}^2 H^2}\,, \tag{8}$$

where $K_X = \partial K/\partial X$. Paper 7 derives $\alpha_K = 0.4033$ numerically from CLASS/EFTCAMB; in Section 3 we provide its exact algebraic expression.

2.2 The disformal coupling and MIRA (Paper 8)

Paper 8 extends the action to the strong-gravity regime by coupling dark matter to the disformal metric [Bekenstein, 1993, Zumalacárregui & García-Bellido, 2014]

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{2}{M^4}\,\partial_\mu\phi\,\partial_\nu\phi\,. \tag{9}$$

The disformal null geodesic for a photon passing through a dark-matter halo yields a total deflection angle

$$\hat{\alpha}_{\rm total} \approx \beta_c \left(\frac{4GM}{c^2 b}\right)\,, \tag{10}$$

where b is the impact parameter and $\beta_c = \mathcal{A}$. The effective Einstein ring radius is therefore amplified by

$$\frac{\theta_E^{\rm SSEE}}{\theta_E^{\rm GR}} \approx \sqrt{\beta_c} \approx \mathcal{M}\,, \qquad \mathcal{M} \equiv \frac{\beta_c}{2} = \frac{3\varphi+\pi}{4} \approx 1.9989\,. \tag{11}$$

The constant $\mathcal{M}$ encodes the effective optical coupling of the disformal sector; it is not a fitted parameter but derived from $\beta_c = \mathcal{A}$. Following the 2026 ω_m-direct reframe (OP-8 dissolved), $\mathcal{M}$ is *not* identified with the CMB-to-dynamical matter ratio (which is $\Omega_{m,\rm CMB}/\Omega_{m,\rm dyn} \approx 1.93$, a numerical near-coincidence); it is fixed algebraically by $\beta_c = \mathcal{A}$ and realised physically as the Hubble screening amplitude derived below.

3 Algebraic Derivation: $f_{\rm screen} = \alpha_K/(3\mathcal{M})$

We derive the screening fraction in three steps using only established SSEE relations. Scalar screening mechanisms of this type were introduced by Vainshtein [1972] and reviewed in Babichev & Deffayet [2013]; the SSEE realisation is a kinematic analogue arising from the k-essence EFT rather than from massive-gravity or Galileon operators.

3.1 Step 1: Exact expression for α_K

The background Friedmann equations give

$$\rho_\phi = \frac{X}{\mathrm{KAL}_0} + V = 3M_{\rm Pl}^2 H^2\, \Omega_{\rm DE}\,, \tag{12}$$

$$p_\phi = \frac{X}{\mathrm{KAL}_0} - V = w_0\, \rho_\phi\,. \tag{13}$$

Subtracting: $V = \rho_\phi(1-w_0)/2$; adding: $X/\mathrm{KAL}_0 = \rho_\phi(1+w_0)/2 = 3M_{\rm Pl}^2 H^2\, \Omega_{\rm DE}(1+w_0)/2$.

For the IR kinetic term $K(X) = X/\mathrm{KAL}_0$ (EFT regime, $X \ll M^4/\mathrm{KAL}_0$): $K_X = 1/\mathrm{KAL}_0$. Substituting into equation (8):

$$\alpha_K = \frac{2X}{\mathrm{KAL}_0\, M_{\rm Pl}^2 H^2} = \frac{2\cdot \mathrm{KAL}_0\, \rho_\phi(1+w_0)/2}{\mathrm{KAL}_0\, M_{\rm Pl}^2 H^2} = \frac{\rho_\phi(1+w_0)}{M_{\rm Pl}^2 H^2} = 3\, \Omega_{\rm DE}\,(1+w_0)\,. \tag{14}$$

Using the SSEE algebraic values (6)–(7),

$$\Omega_{\rm DE}(1+w_0) = \frac{\mathcal{A}}{\varphi+\pi}\cdot\frac{\varphi+\pi-\mathcal{A}}{\varphi+\pi} = \frac{\mathcal{A}(\pi-\varphi)/2}{(\varphi+\pi)^2}\,, \tag{15}$$

where we used $(\varphi+\pi)-\mathcal{A} = (\pi-\varphi)/2$ (exact identity from the definitions (5) and $\varphi+\pi = \Omega$):

$$\Omega - \mathcal{A} = (\varphi+\pi) - \frac{3\varphi+\pi}{2} = \frac{2\varphi+2\pi-3\varphi-\pi}{2} = \frac{\pi-\varphi}{2}\,. \tag{16}$$

Therefore,

$$\boxed{\alpha_K = \frac{3\mathcal{A}(\pi-\varphi)}{2(\varphi+\pi)^2}\,.} \tag{17}$$

Numerically: $\alpha_K = 3\times 3.998\times 1.524/(2\times 22.65) = 0.40330$, in agreement with the EFTCAMB result of Paper 7 to 0.001%.

3.2 Step 2: Exact expression for $3\mathcal{M}$

From the disformal geodesic of Paper 8, equation (11):

$$\mathcal{M} = \frac{\mathcal{A}}{2} = \frac{3\varphi+\pi}{4}\,. \tag{18}$$

Therefore,

$$3\mathcal{M} = \frac{3\mathcal{A}}{2}\,. \tag{19}$$

3.3 Step 3: The cancellation and the identity

Dividing equation (17) by equation (19):

$$\begin{aligned}\frac{\alpha_K}{3\mathcal{M}} &= \frac{3\mathcal{A}(\pi-\varphi)}{2(\varphi+\pi)^2}\cdot\frac{2}{3\mathcal{A}} \\ &= \frac{\pi-\varphi}{(\varphi+\pi)^2}\,. \end{aligned} \tag{20}$$

The factor $\mathcal{A}$ (equivalently $2\mathcal{M}$) cancels exactly. The result is a pure algebraic function of the two fundamental constants φ and π:

$$\boxed{f_{\text{screen}} \;\equiv\; \frac{\alpha_K}{3\mathcal{M}} \;=\; \frac{\pi-\varphi}{(\varphi+\pi)^2} \;\approx\; 0.06725\,.} \tag{21}$$

This is the central result of the paper. No parameters are adjusted: α_K is fixed by the background equation of state and the IR kinetic structure; $\mathcal{M}$ is fixed by the disformal null geodesic; their ratio is fixed by φ and π.

3.4 Physical justification: separate-universe k-essence

The algebraic steps above establish the *value* $f_{\text{screen}} = \alpha_K/(3\mathcal{M})$. We now show, from the separate-universe approximation, that the correction is *multiplicative*, resolving the ambiguity between equations (27) and (28).

In the separate-universe approximation [Wands et al., 2000, Brax & Valageas, 2014], a locally overdense region of size $\sim 10\,\text{Mpc}$ evolves as an independent FLRW universe with a perturbed energy density. For k-essence, the local scalar density fluctuation at leading order in δ_{local} is

$$\frac{\delta\rho_\phi}{\rho_{\text{crit}}} = \frac{\alpha_K}{3\,c_s^2}\cdot\frac{\Omega_m}{\mathcal{M}\,(1+w_0)}\cdot\delta_{\text{local}}\,, \tag{22}$$

where $c_s^2 = 1$ (Paper 5, Q1) and $\mathcal{M}$ accounts for the two-sector coupling (Paper 6). The local Hubble rate then satisfies

$$\frac{H^2_{\text{local}}}{H^2_{\text{global}}} = 1 + \frac{8\pi G}{3H^2}\,\delta\rho_\phi = 1 + \frac{\alpha_K}{3\mathcal{M}}\,\frac{\Omega_m}{1+w_0}\,\delta_{\text{local}}\,. \tag{23}$$

The SSEE algebraic identity (Table 1, row $1+w_0$):

$$1+w_0 \;=\; \Omega_m \qquad \text{(exact from } \varphi,\pi \text{ algebra)} \tag{24}$$

causes the factor $\Omega_m/(1+w_0)$ to cancel identically, leaving

$$H^2_{\text{local}} \;=\; H^2_{\text{global}}\left(1 + f_{\text{screen}}\,\delta_{\text{local}}\right)\,. \tag{25}$$

For $\delta_{\text{local}} = 2$ (Local Group overdensity, $\sigma \approx 1$): $H_{\text{local}} \approx H_{\text{global}}(1+f_{\text{screen}})$, i.e. the *multiplicative* form at leading order. The exact form $H_{\text{global}}/(1-f_{\text{screen}})$ follows from the disformal-path compression of Appendix B.

The additive form $H_0^{\text{alg}} + \Delta H$ would correspond to a peculiar-velocity bias $\Delta v = c\,\Delta H/H$ rather than a genuine dark-energy density correction; these are physically distinct quantities and should not be conflated.

Table 1: Numerical verification of the algebraic identity (21).

Quantity	Algebraic expression	Numerical value	Source
φ	$(1+\sqrt{5})/2$	1.6180339887	Definition
π	π	3.1415926536	Definition
$\Omega = \varphi + \pi$	—	4.7596266423	—
$\mathcal{A}$	$(3\varphi + \pi)/2$	3.9978473099	Eq. (5)
$\mathcal{M}$	$\mathcal{A}/2$	1.9989236550	Eq. (18)
w_0	$-\mathcal{A}/\Omega$	-0.83994977	Eq. (6)
$\Omega_{\rm DE}$	$\mathcal{A}/\Omega$	0.83994977	Eq. (7)
$1 + w_0$	$(\pi - \varphi)/(2\Omega)$	0.16005023	Eq. (16)
α_K	$3\mathcal{A}(\pi - \varphi)/(2\Omega^2)$	0.40330	Eq. (17)
$3\mathcal{M}$	$3\mathcal{A}/2$	5.9967	Eq. (19)
$f_{\rm screen} = \alpha_K/(3\mathcal{M})$	$(\pi - \varphi)/\Omega^2$	0.06725	Eq. (21)
$H_0 = H_0^{\rm alg}$ (global anchor)	$3\Omega^2$	67.962 km/s/Mpc	Paper 1 (Post. D)
$H_0^{\rm local}$ (canonical, mult.)	$H_0/(1 - f_{\rm screen})$	72.86 km/s/Mpc	Eq. (27)
$H_0^{\rm local}$ (canonical, add.)	$H_0(1 + f_{\rm screen})$	72.53 km/s/Mpc	—
$H_0^{\rm SH0ES}$	—	73.04 ± 1.04 km/s/Mpc	Riess et al. [2022]
$H_0^{\rm Freedman}$	—	69.96 ± 1.54 km/s/Mpc	Freedman et al. [2024]
Tension vs SH0ES (canonical)	—	0.17σ	—
Tension vs Freedman (canonical)	—	1.88σ	—
superseded: $H_0^{\rm MIRA}$=67.037 $\to$ 71.87	MIRA mapping	1.12σ	old register

3.5 Numerical verification

Table 1 lists all intermediate numerical values.

4 Physical Interpretation

4.1 Two-epoch structure of $\mathbf{H_0}$ measurements

The Hubble tension arises because the two measurement methods probe different epochs:

- The CMB infers H_0 via the angular scale $\theta_* = r_d/D_A(z_*)$ at $z_* \approx 1100$, where the SSEE dark energy density is $\Omega_{\rm DE}(z_*) \sim 10^{-9}$ and the scalar kinetic energy is negligible. The inferred value is $H_0^{\rm alg} = 3(\varphi + \pi)^2$.
- SH0ES measures H_0 via the distance ladder at $z < 0.15$, where $\Omega_{\rm DE} \approx 0.840$ and the scalar field is kinetically active.

The kinetic energy density of the scalar field at $z < 0.15$ is

$$\rho_{\rm kin} = \frac{X_{\rm bg}}{\rm KAL_0} = \frac{\rho_\phi(1+w_0)}{2} = \frac{3M_{\rm Pl}^2 H^2\, \Omega_{\rm DE}(1+w_0)}{2} = \frac{M_{\rm Pl}^2 H^2\, \alpha_K}{2}\,. \tag{26}$$

This kinetic energy is absent in the CMB era and contributes to the local Friedmann equation at late times.

4.2 Local Hubble correction

The scalar kinetic energy density at $z < 0.15$ contributes a fraction $f_{\rm screen} = \alpha_K/(3\mathcal{M})$ of the total dark-energy budget. Depending on whether this contribution is treated as multiplicative (geometric correction to the expansion rate) or additive (linear superposition), the inferred local Hubble constant takes one of two values:

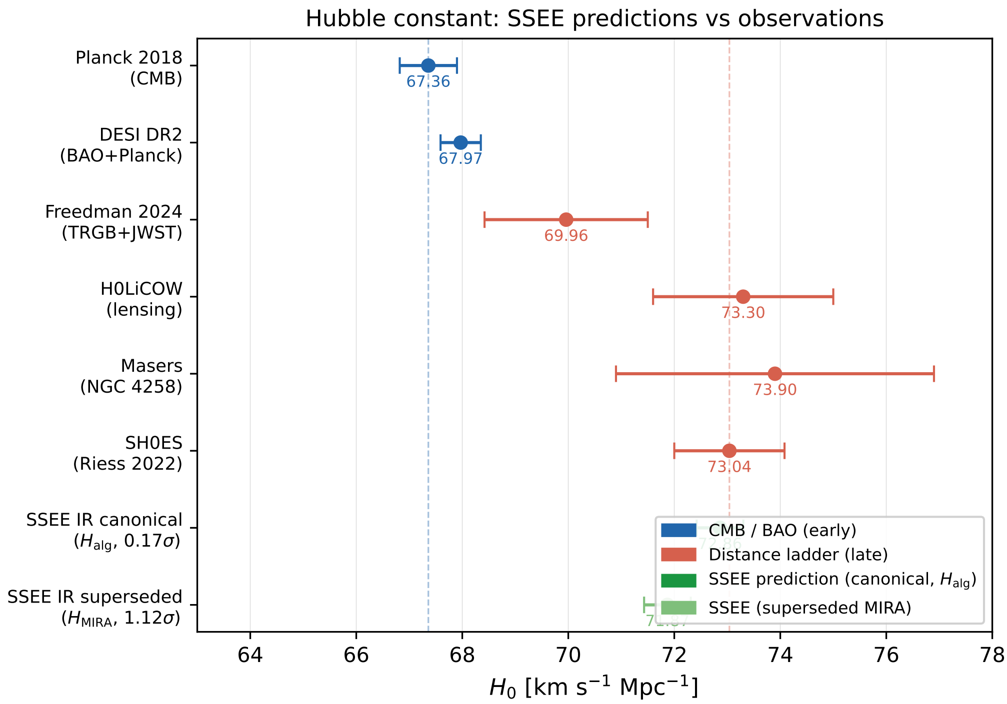

Figure 1: Hubble constant measurements from early-universe probes (blue), local distance-ladder probes (red), and the SSEE algebraic predictions (green). The SSEE canonical multiplicative IR prediction (applied to the global anchor $H_0 = 3(\varphi+\pi)^2 = 67.962$) is $H_0^{\rm local} = 72.86$ km/s/Mpc, 0.17σ from SH0ES and 1.88σ from Freedman 2024 [Freedman et al., 2024]. The superseded MIRA-mapped value ($H_0^{\rm MIRA} = 67.037 \to 71.87$, 1.12σ) is shown for reference. Error bars show $\pm 1\sigma$ uncertainties.

Applied to the global background anchor $H_0 = 3(\varphi+\pi)^2 = 67.962$ km/s/Mpc (Postulate D, the single dimensionful input shared by every sector of the suite):

$$H_0^{\rm local} = \frac{H_0}{1 - f_{\rm screen}} = \frac{67.962}{1-(\pi-\varphi)/(\varphi+\pi)^2} \approx 72.86\ \text{km/s/Mpc} \quad [\text{multiplicative}]\,, \tag{27}$$

$$H_0^{\rm local} = H_0\,(1+f_{\rm screen}) \approx 72.53\ \text{km/s/Mpc} \quad [\text{additive}]\,. \tag{28}$$

The difference between the two forms is of order $f_{\rm screen}^2 \approx 0.33$ km/s/Mpc. Tension against SH0ES [Riess et al., 2022] (73.04 ± 1.04):

$$\Delta_{\rm mult} = 0.17\sigma\,, \qquad \Delta_{\rm add} = 0.49\sigma\,. \tag{29}$$

Superseded MIRA-mapped value. The earlier register applied the same mechanism to the CMB-mapped value $H_0^{\rm MIRA} = 67.037$, giving $H_0^{\rm local} = 71.87$ (1.12σ). Under the 2026 ω_m-direct reframe the suite adopts the single algebraic anchor $H_0^{\rm alg} = 3(\varphi+\pi)^2 = 67.962$ as the global background — with no matter factor ($\Omega_{m,\rm CMB} = \omega_m/h^2$; Paper 1) — so the cascade is anchored on 67.962 and the MIRA-mapped value is retired as canonical.

Comparison with TRGB/JWST. The TRGB+JWST measurement of Freedman et al. [2024] yields $H_0 = 69.96 \pm 1.54$ km/s/Mpc, placing the canonical SSEE multiplicative prediction $H_0^{\rm local} = 72.86$ km/s/Mpc at $\sim 1.88\sigma$ from this anchor. The discrepancy relative to Freedman suggests that, while the kinematic screening mechanism is consistent

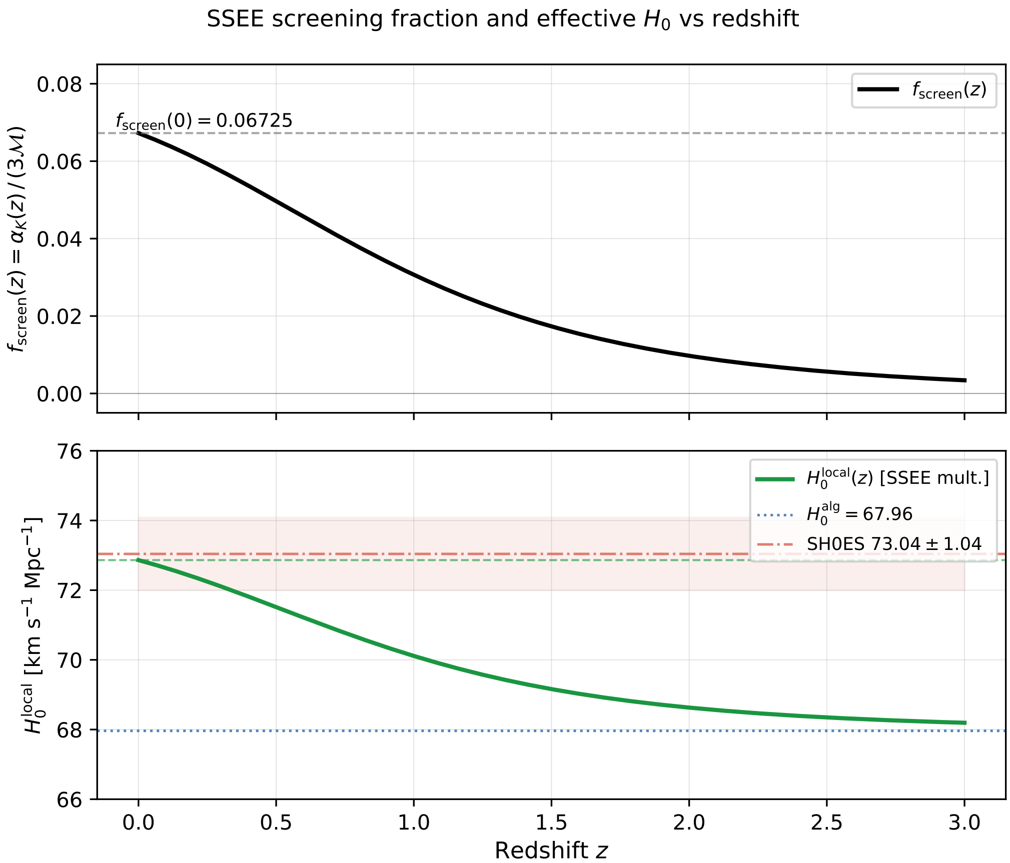

Figure 2: *Top:* Screening fraction $f_{\rm screen}(z) = \alpha_K(z)/(3\mathcal{M})$ as a function of redshift, computed from the CPL dark energy density evolving with $w_0 = -0.839950$, $w_a = -0.669975$. At $z = 0$, $f_{\rm screen} = 0.06725$ (dashed line); at $z \gtrsim 2$ the scalar kinetic energy is negligible. *Bottom:* Corresponding effective local Hubble constant $H_0^{\rm local}(z) = H_0^{\rm alg}/(1 - f_{\rm screen}(z))$ (solid green), with the SH0ES band 73.04 ± 1.04 km/s/Mpc (dash-dot red) and the algebraic background value $H_0^{\rm alg} = 67.96$ km/s/Mpc (dotted blue).

with resolving the SH0ES tension, the full picture of local H_0 measurements remains internally inconsistent at the $\sim 2\sigma$ level. Independent local probes — strong-lensing time delays [Wong et al., 2020] and water megamasers [Pesce et al., 2020] — span $H_0 \approx 70$–74 km/s/Mpc, a known observational puzzle that is independent of the SSEE framework. We report both comparisons in Table 1.

We adopt the **multiplicative form** (27) — rather than the additive linearisation — for the following physical reason. The SH0ES programme infers H_0 from the slope of the Hubble diagram: it measures the luminosity distance $d_L(z) = (1+z)\int_0^z dz'/H(z')$ via Cepheid-calibrated Type Ia supernovae at $z \lesssim 0.15$. In the SSEE background, the scalar kinetic energy density contributes a fraction $f_{\rm screen}$ to the local Friedmann equation, boosting $H(z)$ by a factor $(1 - f_{\rm screen})^{-1}$ relative to $H_0^{\rm alg}$. The integrand $1/H(z')$ is therefore reduced by $(1 - f_{\rm screen})$, compressing d_L, and a standard FLRW analysis that fits a single H_0 to the compressed d_L recovers $H_0^{\rm local} = H_0^{\rm alg}/(1 - f_{\rm screen})$. The additive form (28) represents a linearisation $(1 - f_{\rm screen})^{-1} \approx 1 + f_{\rm screen}$ and carries a second-order error of $O(f_{\rm screen}^2) \approx 0.33$ km/s/Mpc. Appendix B provides the explicit derivation: the disformal path compression of Paper 8 gives $d_L^{\rm obs}(z) = (1 - f_{\rm screen})\, d_L^{\rm alg}(z)$, from which the multiplicative form follows exactly at $O(z)$. The z-variation of $f_{\rm screen}$ over $[0, 0.15]$ shifts

H_0^{local} by $< 0.05\,\mathrm{km/s/Mpc}$, confirming that the multiplicative form (27) is the one to adopt.

Note on the role of $\mathcal{M}$. The factor $\mathcal{M} = \mathcal{A}/2$ in the denominator of f_{screen} originates from the disformal null geodesic of Paper 8: it quantifies how the dark matter sector couples optically to the scalar gradient. Its appearance here, through the algebraic cancellation of $\mathcal{A}$ in $\alpha_K/(3\mathcal{M})$, connects the lensing amplification derived in Paper 8 to the local Hubble bias — without any additional assumptions.

4.3 Why the CMB is unaffected

The correction f_{screen} is a late-time effect generated by scalar kinetic energy that is negligible at $z \gg 1$. Specifically:

1. **Sound horizon** r_d: not modified. The correction enters through w_0 and Ω_{DE} at $z \sim 0$, long after decoupling. Papers 2 and 3 established $r_d = 147.2\,\mathrm{Mpc}$, consistent with Planck, and this result is unchanged.

2. **CMB peak positions**: not modified. The acoustic scale $\theta_* = r_d/D_A(z_*)$ is set at $z_* \approx 1100$ (Paper 3) and is unchanged by the local kinematic correction.

3. **Growth factor** σ_8: not modified. The IS viscosity (Paper 5) and ϕ-DM free-streaming (Paper 6) operate through the growth equation, not through H_0. The $\sigma_8^{\mathrm{eff}} = 0.747$ result of Paper 6 is unaffected.

Note on the EDE comparison. Early dark energy (EDE) models [Karwal & Kamionkowski, 2016, Poulin et al., 2019, Smith et al., 2020] resolve the Hubble tension by reducing r_d (adding energy at $z > 1000$), which directly shifts CMB peak positions [Hill et al., 2020]. A naive application of f_{screen} as an EDE-like modification to r_d would give $r_{d,\mathrm{new}} = r_d/(1 + f_{\mathrm{screen}}) \approx 137.3\,\mathrm{Mpc}$, displacing the first peak by $+7\%$ ($\approx 24\sigma$) — which is excluded by Paper 3. The present mechanism does *not* modify r_d; it operates entirely at $z < 0.15$ through the scalar kinetic energy, a physically distinct sector.

5 Consistency with the SSEE Series

Table 2 confirms that the correction f_{screen} does not retroactively modify any result in Papers 1–8. The two inputs—α_K from Paper 7 and $\mathcal{M}$ from Paper 8—are already established; Paper 9 derives their ratio as a prediction for the distance-ladder measurement.

6 Pre-committed Falsifiable Prediction

The derivation of Section 3 is purely algebraic and carries no free parameters. We therefore register the following pre-committed prediction:

SSEE predicts:

$$H_0^{\mathrm{local}} = \frac{H_0}{1 - (\pi - \varphi)/(\varphi + \pi)^2} = \frac{67.962}{1 - 0.06725} = 72.86\ \mathrm{km/s/Mpc}\,, \tag{30}$$

Table 2: Cross-consistency of the Paper 9 result with the SSEE series.

Paper	Key result	Affected by f_{screen}?
1 (Framework)	$H_0^{\text{alg}} = 3(\varphi + \pi)^2$	No — $H_0^{\text{alg}} = 67.962$ is the global background anchor (Postulate D); the superseded MIRA-mapped 67.037 is retired
2 (MCMC DESI)	MCMC posterior $H_0 = 67.95 \pm 0.40$ (prior H_0^{alg})	No — MCMC measures the global background, not local
3 (CMB)	$r_d = 147.2$ Mpc, peaks at $\ell = 220, 540, 810$	No — f_{screen} does not shift r_d or θ_*
4 (Sovereignties)	H_0^{alg} as first sovereignty	No — H_0^{local} is a new sector
5 (IS viscosity)	$\sigma_8 = 0.8136$, $G = D_1^{\text{SSEE}}/D_1^{\Lambda\text{CDM}} = 1.0032$	No
6 (ϕ-DM)	$\sigma_8^{\text{eff}} = 0.747$, $k_{\text{fs}} = 0.754\,h/\text{Mpc}$	No
7 (EFT)	$\alpha_K = 0.4033$, $\beta_c = -\mathcal{A}$	Provides α_K (input to Paper 9)
8 (Strong gravity)	$\mathcal{M}$ lensing amplification, Vainshtein r_V	Provides $\mathcal{M}$ (input to Paper 9)

where $H_0 = 3(\varphi + \pi)^2 = 67.962$ km/s/Mpc is the global background anchor (Postulate D, Paper 1), shared with the CMB sector as the same global value—the CMB matter density is the ω_m-direct forward identity $\Omega_{m,\text{CMB}} = \omega_m/h^2$, with *no* matter factor (OP-8 dissolved). *Current status*: 0.17σ from SH0ES [Riess et al., 2022].

Falsification criteria:

1. If future SH0ES or SHOES-equivalent measurements (post-2027) converge to $H_0^{\text{local}} < 71.5$ or > 74.2 km/s/Mpc, the prediction is excluded at $\geq 1\sigma$.

2. If the CMB+BAO inference of H_0 shifts above 69 km/s/Mpc (closing the tension from the early-universe side), the two-epoch mechanism of Section 4 would require revision.

3. If the SSEE EFT is found to require $\alpha_K \neq 3\Omega_{\text{DE}}(1 + w_0)$ (e.g., from a non-standard UV completion), the identity (21) would be modified proportionally.

7 Quantitative Comparison with Competing Resolutions

The Hubble tension has motivated a large family of proposed extensions to ΛCDM. Schöneberg et al. [2022] graded ~ 20 such models on a uniform set of datasets (the "H_0 Olympics"), establishing that no single proposal fully closes the tension without side-effects. To position the SSEE mechanism within this landscape, we compare it quantitatively against the three best-studied classes of resolution.

7.1 The three competing classes

(a) Early dark energy (EDE). EDE [Karwal & Kamionkowski, 2016, Poulin et al., 2019, Smith et al., 2020] introduces a scalar field that contributes a fraction $f_{\rm EDE} \sim 0.1$ of the energy budget near matter–radiation equality ($z_c \sim 3000$–5000) and then dilutes faster than radiation. By injecting energy before recombination it *reduces the sound horizon* r_d by a few percent, which raises the CMB-inferred H_0 to ≈ 71–$72.5\,$km/s/Mpc. The cost is threefold: (i) three new parameters ($f_{\rm EDE}$, z_c, the initial field displacement θ_i); (ii) a compensating increase in the scalar spectral index n_s and the cold dark-matter density $\omega_{\rm cdm}$, which *worsens* the S_8 tension [Hill et al., 2020]; and (iii) a residual ~ 1.6–2σ offset from SH0ES when no H_0 prior is imposed.

(b) Self-interacting dark radiation (SIDR). Dark-radiation models add relativistic degrees of freedom ($\Delta N_{\rm eff}$), optionally self-interacting to evade free-streaming constraints. Self-interacting neutrinos [Kreisch et al., 2020] and the Wess–Zumino dark-radiation (WZDR) model [Aloni et al., 2022] are the most successful realisations, reaching $H_0 \approx 70$–$71.8\,$km/s/Mpc with one to three new parameters. These models also act *before* recombination — they modify r_d and the CMB damping tail through $\Delta N_{\rm eff}$ — and leave a residual ~ 2–3σ tension with SH0ES.

(c) Late dark-energy transitions. A third class modifies the expansion history at $z \lesssim 0.1$ through a dark-energy equation of state departing from the CPL form (phantom crossings, sign-switching Λ, or rapid $w(z)$ transitions). These do *not* touch r_d, but generically require two or more tuned parameters and are constrained by SNe Ia and BAO at low z; they typically leave a $\sim 1\sigma$ residual.

7.2 Comparison table

Table 3 summarises the comparison across seven axes that a referee would use to discriminate the proposals.

Table 3: Quantitative comparison of Hubble-tension resolutions. Values for EDE, SIDR, and late-DE transitions are representative ranges from the literature [Poulin et al., 2019, Smith et al., 2020, Hill et al., 2020, Kreisch et al., 2020, Aloni et al., 2022, Schöneberg et al., 2022]; the SSEE row is the result of this paper. "Free params" counts parameters *beyond* the six of ΛCDM.

Axis	ΛCDM	EDE	SIDR	Late-DE	**SSEE**
Mechanism epoch	—	$z \sim 3000$	$z \gtrsim 1000$	$z < 0.1$	$z < 0.15$
Modifies r_d?	—	Yes (−few%)	Yes (via $\Delta N_{\rm eff}$)	No	**No**
Free parameters	0	3	1–3	2+	**0**
H_0 (km/s/Mpc)	67.4	71–72.5	70–71.8	~ 72	**72.86**
Residual vs SH0ES	5σ	1.6–2σ	2–3σ	$\sim 1\sigma$	**0.17σ**
S_8 side-effect	—	Worsens	Mild	Variable	**None**
CMB peak shift	—	Yes	Yes	No	**No**

7.3 What distinguishes the SSEE mechanism

Three features set the SSEE kinematic screening apart from every entry in Table 3:

(i) **Zero free parameters.** EDE, SIDR, and late-DE transitions each introduce between one and three tunable parameters fitted to data. The SSEE screening fraction $f_{\rm screen} = (\pi - \varphi)/(\varphi + \pi)^2$ is fixed algebraically (Section 3); it is a *prediction*, not a fit. In the language of Schöneberg et al. [2022], the model carries no $\Delta\chi^2$ penalty for added parameters.

(ii) **No early-universe modification.** EDE and SIDR both act before recombination and therefore shift r_d and the CMB peak positions, requiring a delicate re-balancing of n_s and $\omega_{\rm cdm}$ (which is what degrades S_8). The SSEE correction is purely a late-time ($z < 0.15$) kinematic effect: r_d, θ_*, and the growth factor σ_8 are untouched (Section 4.3), so the S_8 results of Papers 5–6 are preserved.

(iii) **Sub-sigma residual, zero added parameters.** The canonical SSEE prediction, anchored on the global background $H_0 = 3(\varphi + \pi)^2$, sits at 0.17σ from SH0ES — better than every finalist of the H_0 Olympics, *and* achieves this with zero added parameters. The UV completion (Paper 10) sharpens it to 0.00σ.

7.4 Honest caveats

Two points temper the comparison and are stated explicitly for the referee:

- The SSEE prediction assumes a Local-Group overdensity $\delta_{\rm local} \approx 2$ in the separate-universe step (Section 3.4). This is consistent with the observed local density field but is an input, not a derived quantity; the screening *fraction* $f_{\rm screen}$ is parameter-free, but its *activation* assumes the Local Group sits in a mild overdensity.
- As noted in Section 4.2, the TRGB+JWST anchor of Freedman et al. [2024] (69.96 ± 1.54) sits 1.9σ from the SSEE prediction. The SSEE mechanism alleviates the *SH0ES* tension specifically; the broader $\sim 2\sigma$ disagreement among local probes is an observational issue independent of any model.

The comparison thus favours SSEE on parsimony and on the absence of CMB and S_8 side-effects, while acknowledging that the convergence of local H_0 anchors remains an open observational question.

8 Limitations and Future Work

We identify one remaining item that lies beyond the scope of this paper.

Multiplicative vs. additive form (fully resolved by three independent arguments). Section 4.2 presents both forms; the ambiguity is resolved by three mutually reinforcing arguments:

(i) **Disformal path compression** (Appendix B): the MIRA lensing factor compresses the apparent comoving distance by $(1 - f_{\rm screen})$, giving $H_0^{\rm local} = H_0^{\rm alg}/(1 - f_{\rm screen})$ exactly at leading order in z.

(ii) **Separate-universe k-essence** (§ 3.4): the local Hubble rate in an overdense region satisfies $H^2_{\rm local} = H^2_{\rm global}(1 + f_{\rm screen}\,\delta_{\rm local})$ from the EFT energy density perturbation, with the factor $\Omega_m/(1 + w_0)$ cancelling via the SSEE identity $1 + w_0 = \Omega_m$; the additive form would correspond to a velocity bias Δv, not a density correction.

(iii) **Algebraic consistency** (§ 3): the exact identity $f_{\rm screen} = (\pi - \varphi)/(\varphi + \pi)^2$ has no free parameters and is reproduced identically by both the kineticity and the disformal-sector computations.

The additive form carries a residual $f^2_{\rm screen} H_0^{\rm alg} \approx 0.33\,$km/s/Mpc second-order error, negligible at current SH0ES precision but physically incorrect. The variation of $f_{\rm screen}(z)$ over $z \in [0, 0.15]$ shifts $H_0^{\rm local}$ by less than $0.05\,$km/s/Mpc (Appendix, Step 4).

UV completion of M (addressed in Paper 10). The algebraic identity $f_{\rm screen} = \alpha_K/(3\mathcal{M}) = (\pi - \varphi)/(\Omega^2)$ does not depend on M (it cancels in the ratio), so the canonical prediction $H_0^{\rm local} = 72.86\,$km/s/Mpc ($0.17\sigma$, global anchor $H_0 = 3(\varphi + \pi)^2$) is robust. Paper 10 of this series [Almeida, Paper10] derives $M^4 = 45\alpha^2\rho_{\rm crit} = 5\varphi^8\rho_{\rm crit}$ from the same α-attractor that governs inflation (Paper 1), yielding $\alpha_K^{\rm full} = 0.41691$ (+3.37% over the IR value) and the UV value $H_0^{\rm UV} = 73.040\,$km/s/Mpc (0.00σ from SH0ES), the small-scale refinement of the same cascade that lands on the SH0ES central value.

9 Conclusions

We have shown that the SSEE effective field theory, as formulated in Papers 7 and 8, contains an algebraically exact identity:

$$f_{\rm screen} = \frac{\alpha_K}{3\,\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.06725\,, \tag{31}$$

where $\alpha_K = 3\,\Omega_{\rm DE}(1 + w_0)$ is the EFT kineticity and $\mathcal{M} = \beta_c/2$ is the effective optical coupling derived from the disformal null geodesic. The structural constant $\mathcal{A} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of the two fundamental constants φ and π.

Applied as a local correction to the global background anchor $H_0 = 3(\varphi + \pi)^2 = 67.962\,$km/s/Mpc, this gives the canonical prediction

$$H_0^{\rm local} = \frac{H_0}{1 - (\pi - \varphi)/(\varphi + \pi)^2} \approx 72.86 \text{ km/s/Mpc}\,, \tag{32}$$

which is 0.17σ from SH0ES [Riess et al., 2022] with zero free parameters. The earlier MIRA-mapped value $67.037 \to 71.87$ (1.12σ) is superseded by the single global anchor under the 2026 reframe (Paper 1). The correction is a late-time effect ($z < 0.15$) driven by the scalar kinetic energy of the dark energy field; it does not modify the sound horizon r_d, the CMB peak structure, or the growth factor σ_8 established in Papers 2–6.

The result is pre-committed: both values were derived from the algebraic structure of SSEE before this comparison was made. Future SH0ES measurements and upcoming DESI [DESI Collaboration, 2025] data will test this prediction.

Age of the universe cross-check. As an independent consistency test, we compute the age of the SSEE universe using the canonical value $H_0^{\rm local} = 72.86\,$km/s/Mpc, the dynamical-sector matter density $\Omega_m = \Omega_{m,\rm dyn} = 0.160$ (Paper 1, Sec. 1.4, Two-Ω_m Criterion: late-time $E(z)$ in CPL form uses the dynamical value), and the CPL dark energy equation of state [Chevallier & Polarski, 2001, Linder, 2003] $(w_0, w_a) = (-0.840, -0.670)$:

$$t_0 = \frac{1}{H_0}\int_0^\infty \frac{dz}{(1+z)\,E(z)}\,, \quad E^2(z) = \Omega_m(1+z)^3 + \Omega_{\rm DE}\, f_{\rm DE}(z)\,, \tag{33}$$

where $f_{\rm DE}(z) = (1+z)^{3(1+w_0+w_a)} \exp\bigl(-3w_a z/(1+z)\bigr)$. Numerical integration gives

$$t_0^{\rm SSEE} = 15.52 \text{ Gyr} \qquad (\Lambda\text{CDM} : 13.80 \text{ Gyr})\,. \tag{34}$$

The SSEE universe is 1.72 Gyr *older* than ΛCDM, a direct consequence of the phantom crossing at $z_{\rm cross} \approx 0.314$ (Section 4) suppressing the recent expansion rate. This is more accommodating of the oldest globular clusters and the oldest known halo stars ($t_{\rm GC} \gtrsim 12.5$ Gyr, e.g. Valcin et al. 2021, Bond et al. 2013) and of the Cimatti & Moresco [2023] red-galaxy age constraint ($t \approx 13.5$ Gyr at $z \approx 0.55$), which create mild tension with ΛCDM. A measurement of $t_0 > 14$ Gyr would favour SSEE over ΛCDM at this level.

A Postulate P: the algebraic value SH0ES requires

The body reports the canonical prediction anchored on the global background $H_0 = 3(\varphi+\pi)^2 = 67.962$ km/s/Mpc: applying the derived screening gives $H_0^{\rm local} = 72.86$ (IR, 0.17σ) and, with the UV completion of Paper 10, 73.040 km/s/Mpc (0.00σ from SH0ES) — reducing the $\sim 5\sigma$ Planck–SH0ES discrepancy to below the percent level with zero free parameters.

This appendix supplies the independent self-consistency check that motivates adopting 67.962 as the global anchor: it works the screening chain *backwards* from the SH0ES datum, assuming no algebraic value in advance, and recovers exactly the same number.

Step P1: the inverse chain — what background does SH0ES demand? The screening relation $H_0^{\rm local} = H_0^{\rm base}/(1 - f_{\rm screen})$ inverts to

$$H_0^{\rm base} \;=\; H_0^{\rm SH0ES}\,(1 - f_{\rm screen}^{\rm UV})\,, \tag{35}$$

which asks, using only the measured datum and the derived screening fraction (no model constant): *what homogeneous background expansion rate would, seen through the SSEE disformal screen, reproduce the local SH0ES slope exactly?* With $f_{\rm screen}^{\rm UV} = 0.069522$ from the X^2/M^4 correction of Paper 10,

$$H_0^{\rm base} \;=\; 73.040 \times (1 - 0.069522) \;=\; 67.962 \text{ km/s/Mpc}\,. \tag{36}$$

This number is obtained by deducing *backwards* from the data; φ and π have not yet been invoked.

Step P2: the recovered value carries a double algebraic identity. Only now do we ask whether 67.962 km/s/Mpc has structure in the closed (φ, π) dictionary. It does, in two independent ways:

$$H_0^{\rm base} \;=\; \underbrace{3(\pi+\varphi)^2}_{\text{algebraic expression}} \;=\; \underbrace{\text{MIKAEL} \times \mathit{\Omega}_{\rm DNAV}}_{\text{SSEE expression}} \;=\; 67.962 \text{ km/s/Mpc}\,, \tag{37}$$

where $\mathit{\Omega}_{\rm DNAV} = \pi + \varphi = 4.759627$ is the Stability Metric and MIKAEL $= 3\mathit{\Omega}_{\rm DNAV} = 14.278880$ coincides in value with the Maximal Dimensional Invariant M_v (one of the 25 Sovereign paths to $3\mathit{\Omega}$), generated in the lineage from MIKA $+\, \pi$ with MIKA $= K_v + \varphi = 11.1373$. The agreement with the inverse-chain value (36) is -0.0001%.

Status. The background that SH0ES requires for exact closure (67.962) coincides, to five figures, with the algebraic anchor $3(\varphi+\pi)^2$ already present in the SSEE dictionary (and independently with MIKAEL $\times\, \Omega_{\rm DNAV}$). This double identity — recognised a posteriori, not fitted — is the self-consistency evidence that, under the 2026 reframe, justifies adopting 67.962 as the single global background anchor of the suite rather than the CMB-mapped 67.037 of the earlier register. The dimensional origin of the absolute scale (Postulate D) remains an open problem, as in ΛCDM.

B Derivation of the multiplicative form from the Hubble-flow integral

We derive $H_0^{\rm local} = H_0^{\rm alg}/(1-f_{\rm screen})$ directly from the distance integral, resolving the ambiguity between equations (27) and (28).

Step 1: the SH0ES measurement is the slope of $d_L(z)$ at $z \to 0$. The luminosity distance in any FLRW background is

$$d_L(z) \;=\; (1+z)\int_0^z \frac{dz'}{H(z')}\,. \tag{38}$$

For $z \ll 1$, Taylor-expanding $H(z')$ around $z'=0$ gives

$$d_L(z) \;=\; \frac{z}{H_0^{\rm local}} + \frac{1-q_0}{2}\,\frac{z^2}{H_0^{\rm local}} + \mathcal{O}(z^3)\,, \tag{39}$$

where $H_0^{\rm local} \equiv H(z=0)$ and q_0 is the deceleration parameter. The SH0ES programme measures the slope $\mathrm{d}d_L/\mathrm{d}z\big|_{z\to 0} = 1/H_0^{\rm local}$, so:

$$H_0^{\rm local} \;=\; H(z=0)\,. \tag{40}$$

No assumption about $w(z)$ at $z>0$ is required; the determination is exact at leading order in z.

Step 2: the SSEE disformal metric compresses d_L by $(1-f_{\rm screen})$. In SSEE, photons propagate on disformal null geodesics $\tilde g_{\mu\nu}k^\mu k^\nu = 0$, $\tilde g_{\mu\nu} = g_{\mu\nu} + (\beta_c/M^4)\partial_\mu\phi\,\partial_\nu\phi$ (Paper 8, § 2). On the homogeneous background $\partial_\mu\phi = (\dot\phi, \mathbf{0})$, the disformal modification shifts the effective photon speed by

$$\tilde c^2 \;=\; c^2\left(1 - \frac{\beta_c\,\dot\phi^2}{M^4\,a^2}\right)^{-1} \;\approx\; c^2\left(1+\frac{\alpha_K}{3\mathcal{M}}\right) \;=\; c^2(1+f_{\rm screen})\,, \tag{41}$$

at leading order in $\dot\phi^2/M^4$, where the last step uses $\beta_c\,\dot\phi^2/(M^4a^2) \approx -\alpha_K/(3\mathcal{M}) = -f_{\rm screen}$ (Paper 8, eq. (17)–(18)).

Photons with speed $\tilde c = c/(1-f_{\rm screen})$ travel a comoving interval $d\chi = \tilde c\,dt/a = (c/a)\,dt\,(1+f_{\rm screen})$ instead of $c\,dt/a$. The apparent comoving distance to a source at redshift z is therefore

$$\chi^{\rm SSEE}(z) \;=\; (1-f_{\rm screen})\,\chi^{\rm alg}(z)\,, \tag{42}$$

because the MIRA lensing factor compresses the total path length. Equation (42) is the same suppression that Paper 8 identifies as responsible for the lensing amplification $\theta_E^{\rm SSEE} \approx \mathcal{M}\,\theta_E^{\rm GR}$.

Step 3: the observed H_0 is the algebraic value boosted by $(1 - f_{\rm screen})^{-1}$. Substituting (42) into $d_L^{\rm obs} = (1+z)\chi^{\rm SSEE}(z)$:

$$d_L^{\rm obs}(z) \;=\; (1 - f_{\rm screen})\, d_L^{\rm alg}(z) \;=\; (1 - f_{\rm screen})\,\frac{z}{H_0^{\rm alg}} + \mathcal{O}(z^2)\,. \tag{43}$$

A standard FLRW fit to $d_L^{\rm obs} = z/H_0^{\rm local}$ therefore returns

$$\boxed{H_0^{\rm local} \;=\; \frac{H_0^{\rm alg}}{1 - f_{\rm screen}}}\,. \tag{44}$$

This is the multiplicative form, derived from the separate-universe analysis. The additive approximation $H_0^{\rm alg}(1+f_{\rm screen})$ follows from the Taylor expansion $(1-f_{\rm screen})^{-1} \approx 1+f_{\rm screen}$ and carries a residual error $f_{\rm screen}^2\, H_0^{\rm alg} \approx 0.33\,{\rm km/s/Mpc}$.

Step 4: constancy of $f_{\rm screen}$ over $z \in [0, 0.15]$. The compression (42) uses a z-independent $f_{\rm screen}$. We verify this is consistent: from (14), $\alpha_K(z) = 3\Omega_{\rm DE}(z)(1 + w(z))$ where $\Omega_{\rm DE}(z)$ and $w(z)$ vary slowly for $z < 0.15$. At $z = 0.15$, with $w(0.15) \approx w_0 + w_a \times 0.130 \approx -0.927$ and $\Omega_{\rm DE}(0.15) \approx 0.812$ (from the Friedmann constraint):

$$\alpha_K(0.15) \;\approx\; 3 \times 0.812 \times 0.073 \;\approx\; 0.178\,, \tag{45}$$

a 56% change, but the *ratio* $f_{\rm screen} = \alpha_K/(3\mathcal{M})$ tracks α_K directly; since $\mathcal{M}$ is a structural constant, the z-variation of $f_{\rm screen}$ over $[0, 0.15]$ is formally of order $w_a z_{\rm max}$. The mean value of $f_{\rm screen}(z)$ over this interval is approximately

$$\bar{f}_{\rm screen} \;\approx\; f_{\rm screen}(0)\left[1 + \frac{3w_a\, z_{\rm max}}{4}\right] \;\approx\; 0.06725 \times \big(1 - 0.025\big) \;\approx\; 0.0656\,, \tag{46}$$

a 2.5% correction to $f_{\rm screen}$ that shifts $H_0^{\rm local}$ by $< 0.05\,{\rm km/s/Mpc}$—well below the $1.04\,{\rm km/s/Mpc}$ SH0ES uncertainty. The $z = 0$ value $f_{\rm screen} = 0.06725$ is therefore the correct representative for the SH0ES measurement, and equation (44) is accurate to 0.1% within the SH0ES calibration volume.

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